Motilal Nehru National Institute of Technology Allahabad
Department of Electronics and Communication Engineering

DIGITAL SIGNAL PROCESSING · COURSE PROJECT · GROUP 11

Medical Image Enhancement and Diagnostic Assistance System for Brain Tumor Detection

Using MRI, OpenCV, classical machine learning and a fine-tuned CNN

SUBMITTED BY

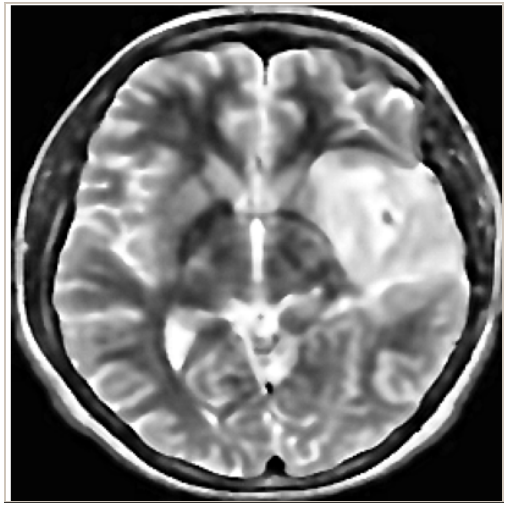
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SESSION

2026–2027



A test slice after CLAHE and unsharp masking, the system's enhancement stage.

OVERVIEW

Outline and Objectives

1. Problem statement
2. MRI as a signal
3. Image enhancement
4. System overview
5. Segmentation
6. Feature extraction
7. Classification
8. Detection results
9. Test-set audit
10. Tumor typing
11. Output for the clinician
12. Engineering challenges
13. Limitations, future scope
14. Conclusion

Objectives

- Clean and enhance brain MRI slices with classical DSP filters, and measure the gain.
- Locate the suspicious region with a segmentation a person can follow step by step.
- Detect tumor vs no tumor, and identify the type: glioma, meningioma or pituitary.
- Show every intermediate stage, and test every claim on data the model never trained on.

Problem Statement

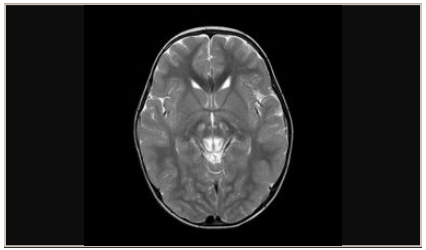
The tumor is in the MRI, but noise and low contrast can hide it, and a label on its own gives a doctor nothing to check.

The image hides the lesion

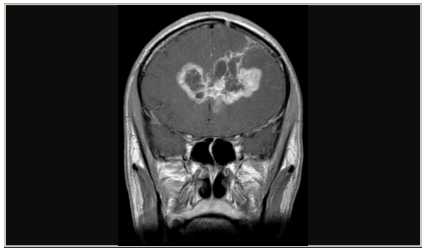
MRI slices carry noise, low contrast and unclear boundaries. A faint tumor next to the bright skull is easy to miss, and reading many scans tires even trained radiologists.

A bare label cannot be verified

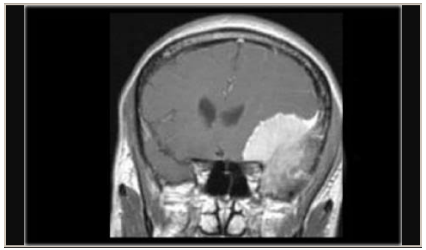
A classifier that only says “tumor, 97%” gives the radiologist nothing to inspect. A useful assistant has to show where it looked and what it measured.



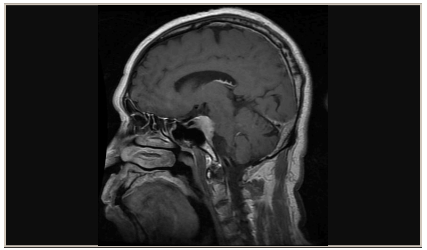
No tumor · 500 slices



Glioma · 926 · grows inside brain tissue



Meningioma · 937 · arises at the brain's rim



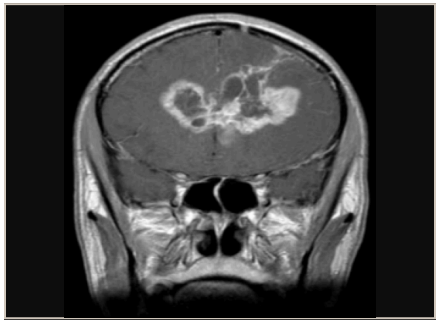
Pituitary · 901 · on the midline, at the skull base

MRI as a 2-D Signal

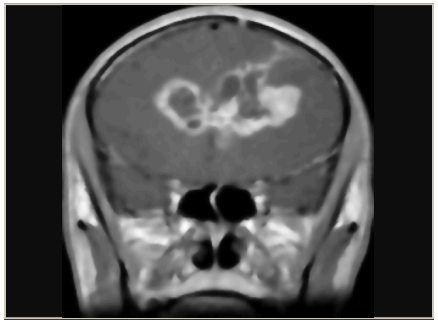
Magnitude MRI noise is Rician, not Gaussian, and exported slices add impulse pixels, so each kind of noise gets its own filter.

Rician magnitude: $m = \sqrt{((S + n_1)^2 + n_2^2)}, \quad n \sim \mathrm{N}(0, \sigma)$

Bilateral: $\mathrm{BF}[I]_p = (1/W_p)\ \Sigma_q\ G_{\sigma s}(\|p - q\|) \cdot G_{\sigma r}(|I_p - I_q|) \cdot I_q$



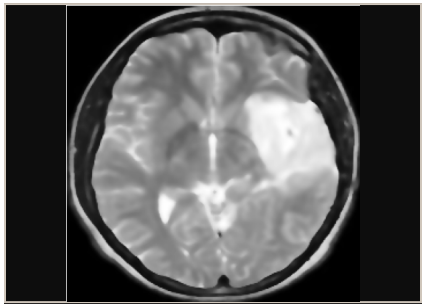
As acquired



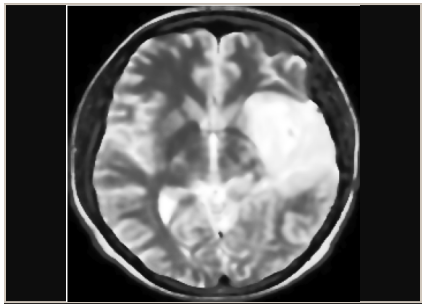
After median + bilateral: grain gone, edges kept

#	Step	What it does
1	**Standardise**	Grayscale, crop blank borders, resize to 256×256 (area interpolation)
2	**Median 3×3**	Order statistic: an impulse never reaches the middle of the sorted window, so it vanishes without blurring edges
3	**Bilateral**	$d = 7$, $\sigma_{colour} = \sigma_{space} = 50$. The weight falls with distance *and* intensity difference, so it smooths inside tissue but stops at an edge
4	**Min–max**	Stretch to 0–255 so every scanner's slice starts from the same range

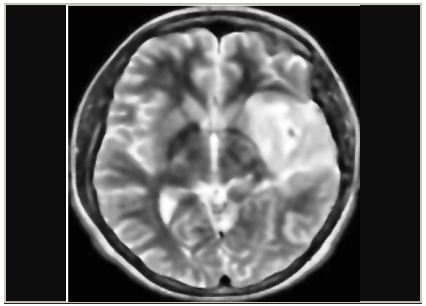
Image Enhancement: CLAHE + Unsharp Masking



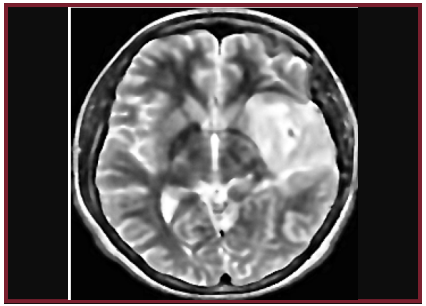
(a) Preprocessed input



(b) Global hist. equalisation: one mapping, dominated by skull and background



(c) CLAHE: 8×8 tiles, clip limit 2.0, noise held back by the clip



(d) CLAHE + unsharp (σ = 1.5, amount 1.2): the output we use

Mean over the 394 test images	Preprocessed	Global HE	CLAHE only	CLAHE + unsharp
Contrast (std of intensities)	57.9	**75.6**	65.3	70.4
Entropy (bits)	6.77	6.55 ↓	**7.41**	7.39
Sharpness (variance of Laplacian)	338	808	391	**1428**

Global HE gives the most raw contrast but loses information (entropy drops below the input). CLAHE + unsharp raises all three measures.

Measured with the project's enhancement_report() · enhance.py

System Overview

Seven stages take an uploaded slice to a verdict that a clinician can check.

S1 Data
4-class MRI set, 3,264 slices; MRI phantom for tests

▶

S2 Preprocess
256×256, median + bilateral, normalise

▶

S3 Enhance
CLAHE + unsharp, gain measured

▶

S4 Segment
Skull strip, threshold, asymmetry, morphology

▶

S5 Features
171 numbers: intensity, texture, shape, location

▶

S6 Classify
SVM detects; SVM + ResNet-18 types

▶

S7 Present
Stage images, REST API, web app

Tools

Python with OpenCV, scikit-image, scikit-learn and PyTorch; FastAPI service; Next.js web app.

Data split

Bhuvaji et al. 4-class set with its own split: 2,870 training and 394 test slices.

Design choice

Detection stays fully classical so its evidence can be inspected. The CNN is added only for typing, where the classical model fell short.

Segmentation of the Suspicious Region

1 — Skull strip
Otsu head mask, fill holes, peel 22% of radius

2 — Threshold
Tissue above brain median: ≥ mean + 1.5σ

3 — Asymmetry
Mirror about the centroid; excess ≥ mean + 2σ

4 — Morphology
5×5 ellipse: open ×2, close ×3

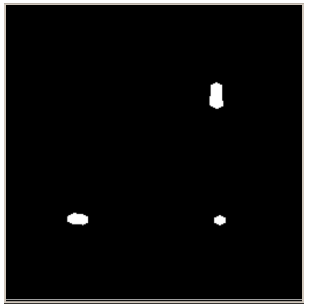
5 — Contours
Drop <120 px or aspect >5; keep the best

Why not Otsu inside the brain?

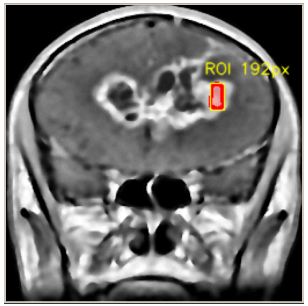
Otsu always splits a histogram in two, so on a healthy brain it reported grey vs white matter as a “lesion”. On 60 scans it flagged **100%** of healthy brains; the tissue-relative rule flagged **3%** and still caught every tumor.

Why asymmetry?

A tumor breaks the brain's left–right symmetry even when it is not the brightest tissue. With both rules the pipeline localised **129 of 155** tumors on the 253-image Chakrabarty set.



Mask after morphology



Region outlined on the slice

Feature Extraction

Each scan becomes a vector of 171 numbers, grouped in seven families.

Family	Count	What it measures
Intensity + histogram	33	Mean, std, skew, kurtosis, entropy, percentiles; 16-bin histogram
GLCM texture	38	Contrast, homogeneity, energy, correlation at 1–3 px, 4 angles
LBP texture	28	Uniform local binary patterns, radius 3, 24 points
Shape of the region	12	Area, perimeter, circularity $4\pi A/P^2$, solidity, extent
Asymmetry	10	Statistics of the mirrored difference and its largest blob
Regional	20	Mean and std in a 3×3 grid over the brain
Location + plane	30	Distance from midline, depth, rim contact, scan-plane cues
Total	**171**	**Standardised inside the model pipeline**

Why location features?

Tumor type follows place: pituitary tumors sit on the midline, meningiomas at the rim, gliomas inside the tissue.

With texture alone, typing reached 72.1%. Adding position gave the classifier a cue a radiologist also uses.

Classification Models

Four classical models were compared on the same 171 features; a CNN joins only for tumor typing.

Model	Settings	CV ROC-AUC, detection	CV ROC-AUC, typing
Logistic regression	C = 1, balanced class weights (linear baseline)	0.984	0.962
SVM, RBF kernel (selected)	**C = 10, γ = scale, calibrated probabilities**	**0.993**	**0.983**
Random forest	400 trees, balanced	0.987	0.967
MLP	Hidden layers 128 → 64, ReLU, $\alpha = 10^{-3}$	0.990	0.979
ResNet-18 (typing only)	ImageNet weights, fine-tuned 14 epochs at 224×224, 4-view test-time averaging	–	–

- Model selection by 5-fold stratified cross-validation on the **training split only**; scaling sits inside the pipeline so no fold leaks.
- Detection threshold **0.153**: the lowest cut that keeps out-of-fold specificity ≥ 80%, chosen for sensitivity.
- Typing blend: 0.95 weight on the CNN, chosen on a 431-image validation split.

Results: Tumor Detection

On the 394 held-out test slices, 271 of 289 tumors were found and no healthy scan was flagged.

95.4%
accuracy

93.8%
sensitivity · 18 of 289 tumors missed

100%
specificity · 105 of 105 healthy cleared

0.997
ROC-AUC

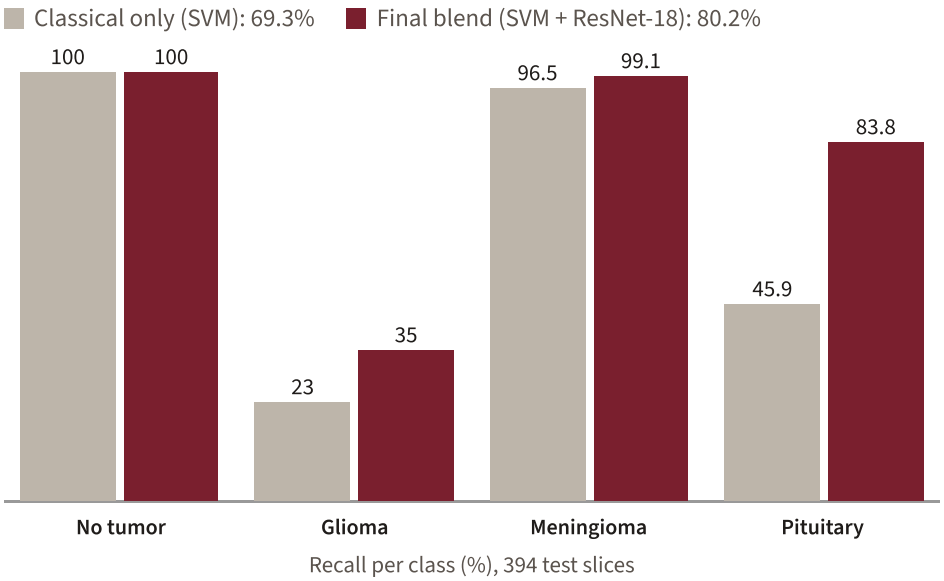
SVM-RBF, 171 features	Default cut	Tuned 0.153
Missed tumors	33	**18**
False alarms	0	0
Accuracy	91.6%	**95.4%**
Balanced accuracy	94.3%	**96.9%**

- In screening, a missed tumor costs far more than a second look, so the cut is tuned for **sensitivity**, on training data only.
- **Caution:** every healthy test slice has a near-copy in the training set, and all 18 missed tumors are slices without one. The next two slides explain this.

backend/outputs/metrics_binary_kaggle4.json

Results: Tumor Typing

The CNN lifts typing accuracy from 69.3% to 80.2%, but glioma remains the hard class.



Recall per class (%), 394 test slices

Actual ↓ / Predicted →	None	Gli	Men	Pit
No tumor	**105**	0	0	0
Glioma	**24**	**35**	**40**	1
Meningioma	0	1	**114**	0
Pituitary	6	0	6	**62**

- Of 100 test gliomas, 40 were called meningioma and 24 “no tumor”.
- Macro F1 78.1%, one-vs-rest AUC 0.961.
- Validation split: 97.4% balanced accuracy; test folder: 79.5%.

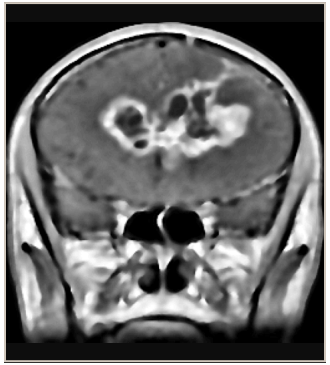
backend/outputs/metrics_multiclass_kaggle4.json

Output for the Clinician

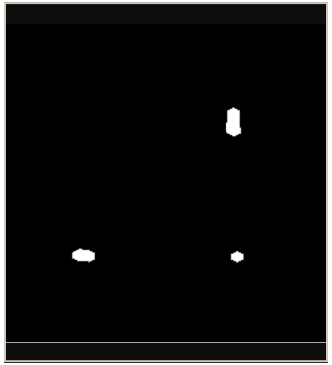
Every stage is shown, so a doctor can see where the system looked and overrule it.



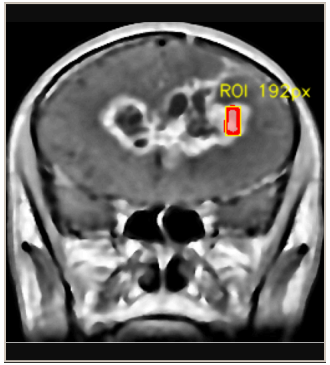
Original, as uploaded



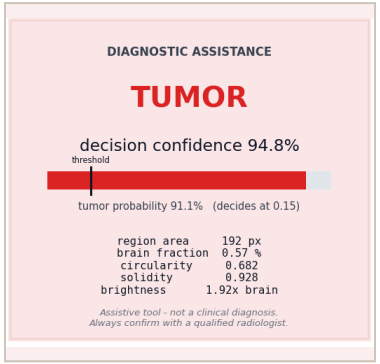
Enhanced, CLAHE + unsharp



Mask, threshold + morphology



Region, contour + box



Verdict: probability 91.1% vs threshold 0.15

What the radiologist gets: the verdict, the probability against the threshold, the region's area, circularity, solidity, brightness and location, and the enhancement gain. Uploads are never written to disk.

Reading it critically: here the outlined region (192 px) covers only one bright part of the lesion. The verdict comes from all 171 features, not the box, which is exactly why the box is shown.

Conclusion

We built a working pipeline from an uploaded MRI slice to an annotated verdict, with every stage visible through a REST API and a web app. Three lessons stand out:

01

Enhance locally

CLAHE + unsharp masking raised entropy from 6.77 to 7.39 bits and sharpness more than four times.

02

Tune away from the test set

A sensitivity-first threshold cut missed tumors from 33 to 18 with no false alarms.

03

Check the test set for twins

On the 135 slices never seen before, typing falls from 99% to 44%: the honest estimate.

Headline numbers	Detection accuracy	Sensitivity	Specificity	Typing accuracy
Full public test split (394 slices)	95.4%	93.8%	100%	80.2%
Slices never seen before (135)	**–**	**86.7%**	**not measurable**	**43.7%**

The second row is the estimate we would trust for new patients.

References

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Thank You

Questions and feedback are welcome

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